Long-Term Impact of Dam Burst on Rapeseed Production in Vasylivskyi, Ukraine

By Kayla Paramore

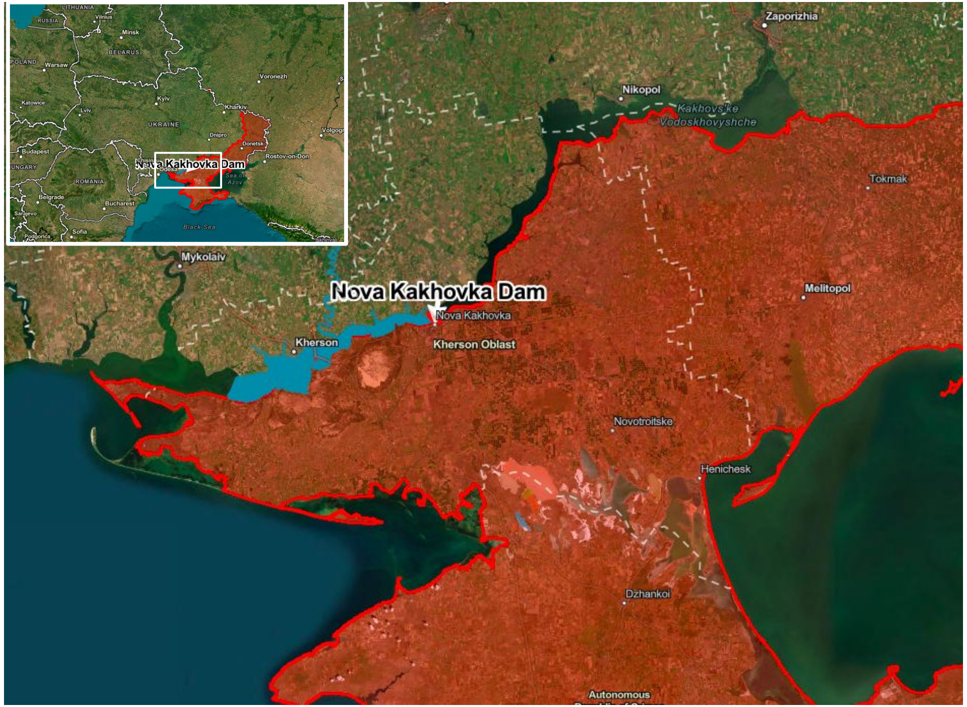
Introduction

A Vital Agricultural Center in a Conflict Zone

- On June 6, 2023, The Nova Kakhovka Dam collapsed, flooding nearby towns and draining upstream canal systems of water

- Ukraine’s was in among the world’s top ten producers of wheat, sunflowers, maize and rapeseed prior to the war.

- Monitoring the effect on agriculture is made more challenging because it is a conflict zone.

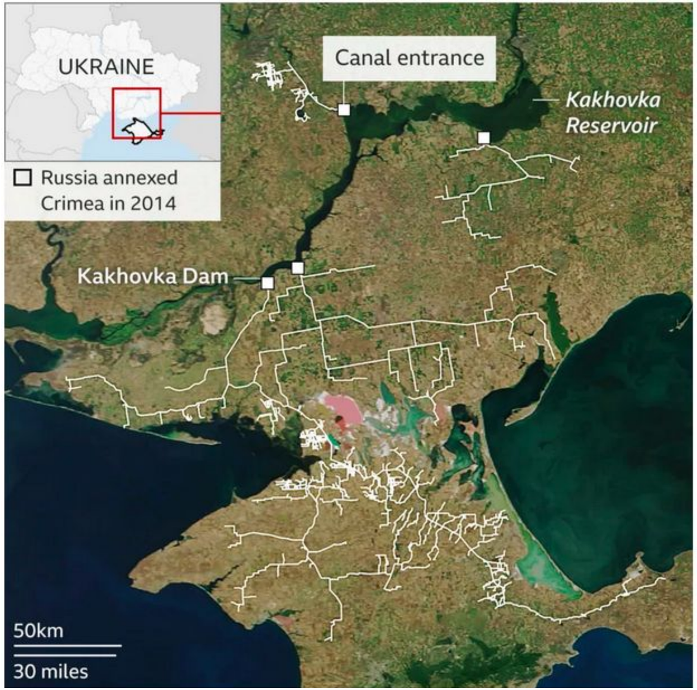


June 2023 Map of Dam Collapse. [Source]

Damage to Ukraine’s Canals Network

- Immediate inundation of towns and fields downstream

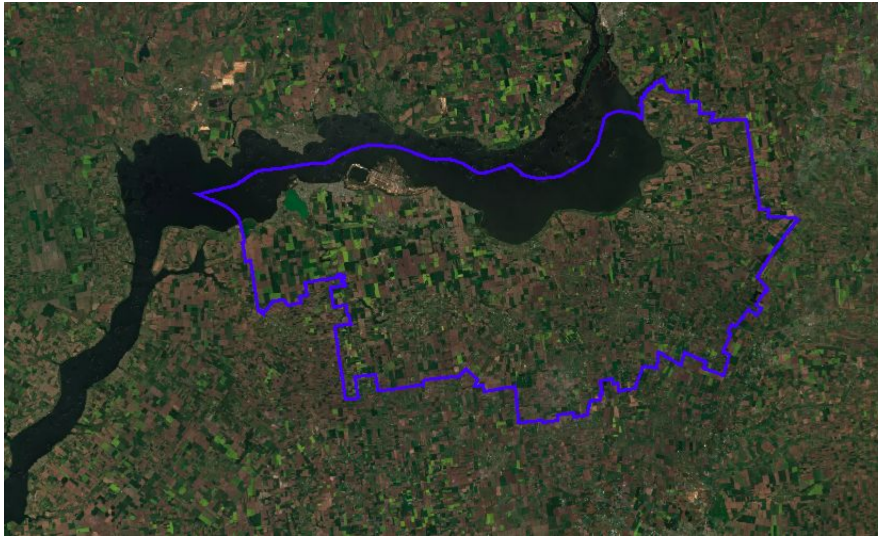
- Canals fed by the dam run dry



Canal network fed by the Dnipro River. [Source]

Study Area: Valsylivskyi District, Ukraine

- Vasylivskyi district
- 4,308 sq km area
- South of the Kakhovka Reservoir
- Part of Ukraine’s agricultural heartland



Focus on Rapeseed (Winter Oilseed) Production

- The rapeseed (*Brassica napus*) is used to make Canola Oil

- It is planted in the fall (late August/September). And Harvested in the summer (July/Aug)

Image Sources: link1, link 2

Distinctive yellow flowers prior to senescence

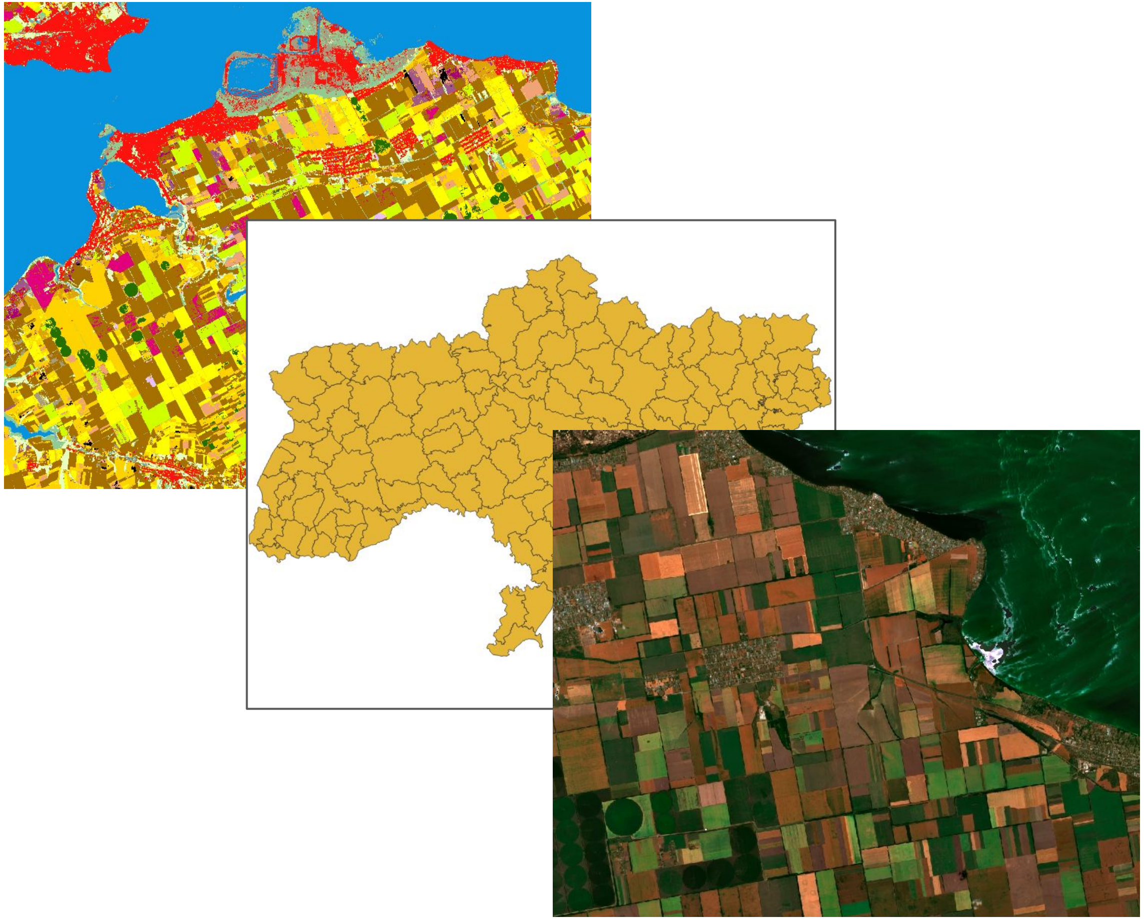
May 2022

June 2022

Image Sources: [link1](), [link 2]()

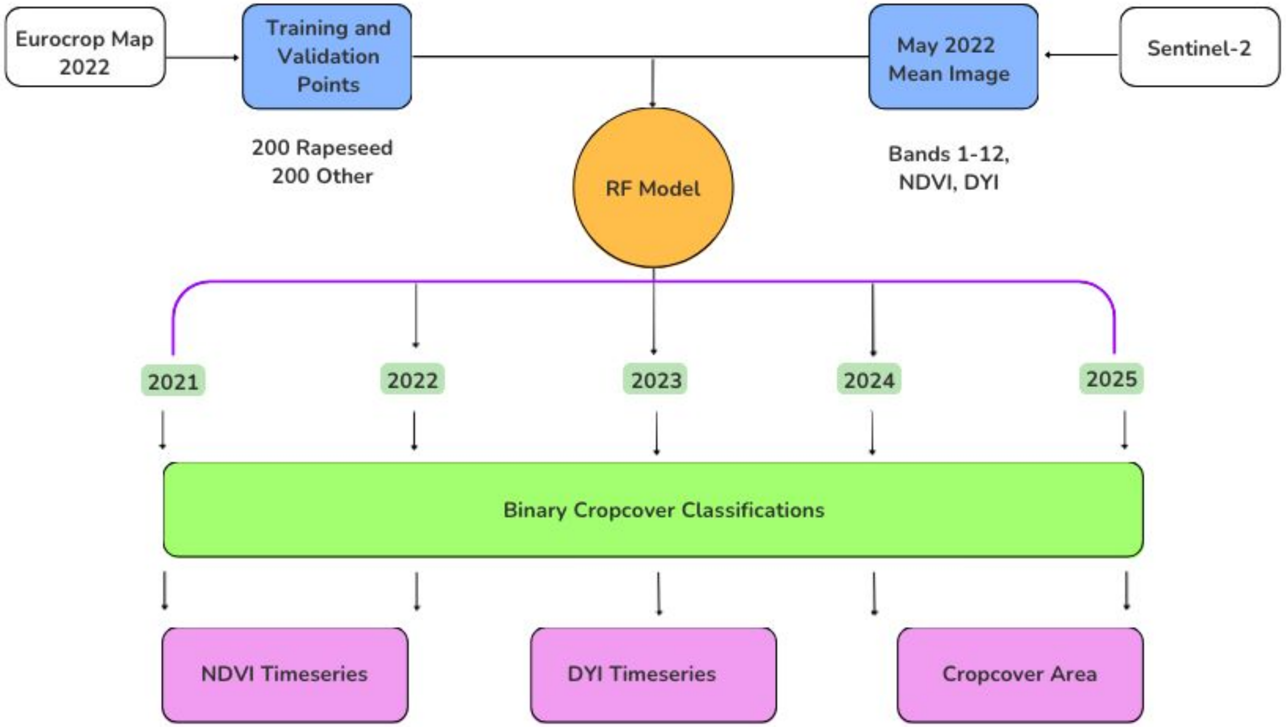
Data

- Eurocrop Map
- Ukraine Boundaries
- Sentinel-2



Methodology

Methodology

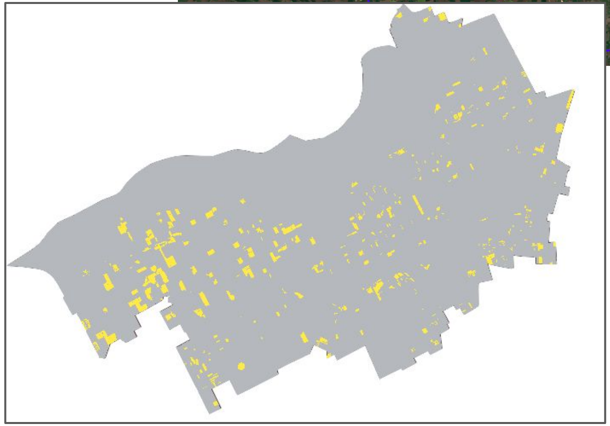


Random Forest Model to Classify Rapeseed Cropcover

- Training points derived from Eurocrop map 2022

- Sentinel-2 B1-B11 bands, plus NDVI and DYI (DYI band used to pick up May crop flowering)

- 200 Rapeseed points, 200 Other points; 50 decision trees



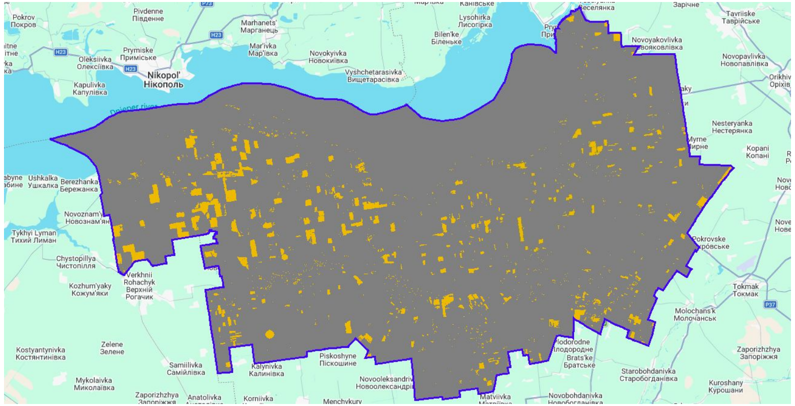
May 2022

Classification of Rapeseed Cropcover + Total Cropcover Area

- Applied the RF Model to the years 2021-2025

- 2022 Classification had a 98% accuracy

- Other years' classifications were not evaluated for accuracy



0: [80,2]
1: [1,83]

Accuracy
0.9819277108433735

Kappa
0.9638449252214317

total samples
400

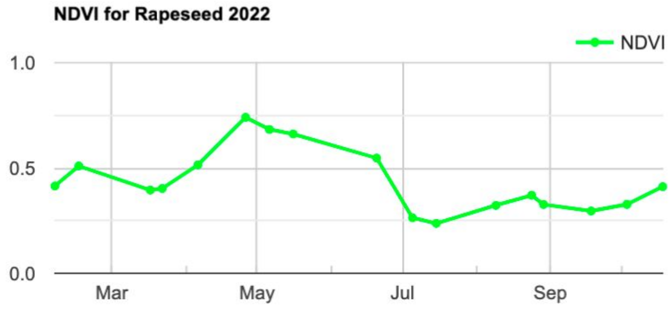
training sample size
234

validation sample size
166

Calculate NDVI and DYI Timeseries Charts for Each Year

- Timeseries show the phenological cycle of the plant

- Uses a single scene, for NDVI consistency

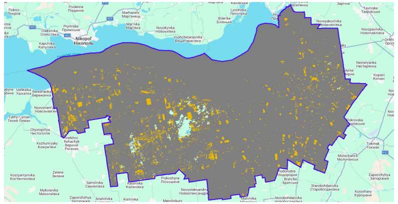


Results

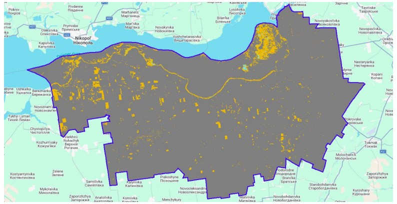
Classification Maps

- Effective, but over-predicts rapeseed, including post-flood water pollution

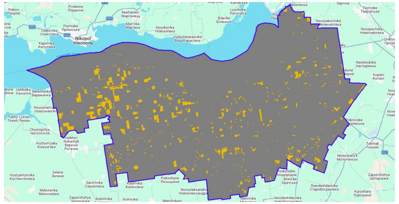
- Given crop rotation practices, rapeseed covers approx. 35,000 ha - 65,000 per year



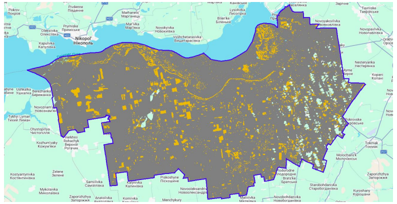
2021



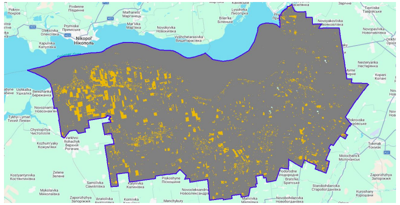
2024



2022



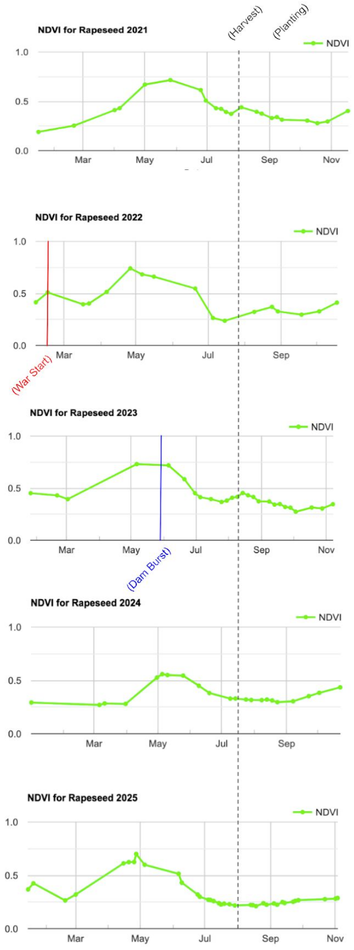
2025



2023

Impact on NDVI

- No visible effect on NDVI in 2023, as crops were already in senescence.
- Pronounced decrease in 2024, following dam burst.
- 2025, NDVI appeared improved, but not returned to pre-dam burst norms



Discussion + Conclusions

Future Work

- Estimate yield based on NDVI

- Improve the Model - particularly, post-flood years

- Creating validation for other years

- Classifying other crop types and running a similar analysis